



Azure Service Bus

Azure Service Bus is a cloud-based messaging service provided by Microsoft Azure that enables reliable communication between applications and services, whether they are running in the cloud, on-premises, or in a hybrid environment. It acts as a message broker, decoupling application components to improve scalability, resilience, and performance.

Key Features:

- **Message Queues:** Enables asynchronous communication by storing messages until they are processed.
- **Topics and Subscriptions:** Supports publish-subscribe messaging, allowing multiple subscribers to receive the same message.
- **Sessions:** Supports message sequencing and grouping for ordered processing.
- **Dead-letter Queues:** Handles undelivered or failed messages for later inspection.
- **Geo-disaster Recovery:** Ensures high availability and failover capabilities across regions.
- **Advanced Security:** Supports Role-Based Access Control (RBAC), Managed Identities, and encryption.

Use Cases:

1. Decoupling Microservices

- In a microservices architecture, Azure Service Bus allows independent services to communicate without direct dependencies, improving scalability and maintainability.

2. Order Processing Systems

- E-commerce platforms use Service Bus to queue orders, ensuring that processing occurs in a reliable and sequential manner.

3. Event-driven Architectures

- Applications that rely on event-driven workflows can use topics and subscriptions to distribute events to multiple consumers.

4. Hybrid Cloud Integration

- Enterprises with on-premises and cloud-based applications use Service Bus to ensure seamless data transfer and synchronization.

5. IoT Data Ingestion

- Devices send telemetry data to Service Bus, allowing backend systems to process messages efficiently.

6. Reliable Transaction Processing

- Financial institutions use Service Bus for secure and ordered transaction processing with retry mechanisms.

Benefits:

- **Scalability:** Handles large message volumes efficiently.
- **Reliability:** Guarantees message delivery with retries and dead-lettering.
- **Security:** Provides enterprise-grade authentication and encryption.
- **Flexibility:** Supports multiple messaging patterns, including queues and publish-subscribe.
- **Integration:** Works seamlessly with Azure services like Functions, Logic Apps, and Event Grid.

Azure Service Bus is ideal for applications requiring robust messaging capabilities with high availability and security. Let me know if you need more details on implementation or pricing.

😊 To begin with the Lab

1. In the marketplace search for Service bus and click on Create.



Service Bus

♥ Add to Favorites

Microsoft | Azure Service

★ 4.2 (263 ratings)

Plan

Service Bus



Create

2. Then choose your resource group and in the namespace give it name then for the pricing tier choose Standard.

Project Details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance Details

Enter required settings for this namespace.

Namespace name * .servicebus.windows.net

Location *

Pricing tier * [Browse the available plans and their features](#)

- In Azure Service bus we can create queues on which we can send and receive messages. Here you can see that our Service bus is created now we will create a queue.

Newbus121
Service Bus Namespace

+ Queue
+ Topic
Refresh
Delete
Give feedback

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Settings
- Entities
- Monitoring
- Automation
- Help

Show data for the last: 1 hour 6 hours 12 hours 1 day 7 days 30 days

Requests

Incoming Requests. (Sum), newbus121 | 0
Successful Requests. (Sum), newbus121 | 0
Server Errors. (Sum), newbus121 | 0
User Errors. (Sum), newbus121 | 0
Throttled Requests. (Sum), newbus121 | 0

- Give a name to your queue and click on create.

Create queue



Service Bus

Name * ⓘ

demoQueue ✓

Max queue size

1 GB ▼

Max delivery count * ⓘ

10

Message time to live ⓘ

Days	Hours	Minutes	Seconds
14	0	0	0

Lock duration ⓘ

Days	Hours	Minutes	Seconds
0	0	1	0

☐ Enable auto-delete on idle queue ⓘ

☐ Enable duplicate detection ⓘ

☐ Enable dead lettering on message expiration ⓘ

☐ Enable partitioning ⓘ

☐ Enable sessions ⓘ

Create

 Give feedback

5. In the entities if you go to queues you will see your queue. Go inside.

Home > Newbus121

Newbus121 | Queues

Service Bus Namespace

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Settings
Entities
Queues
Topics
Monitoring
Automation
Help

Search to filter items by name...

Name	Status	Message count	Active messages	Dead-letter messages	Scheduled messages	Max size	Enable partitioning
demoqueue	Active	0	0	0	0	1024 MB	false

6. Now go to Service Bus Explorer and here we can send message to the queue. Click on it.

demoqueue (Newbus121/demoqueue) | Service Bus Explorer

Service Bus Queue

Search

Overview
Access control (IAM)
Diagnose and solve problems
Service Bus Explorer
Settings
Shared access policies
Properties
Locks
Automation
CLI / PS
Tasks
Export template
Help

Peek Mode Send messages Refresh

Queue (0) Dead-letter (0)

Peek from start Peek next messages Peek with options Re-send selected messages Download selected message body

Showing 0 of 0 messages

Sequence Number	Message ID	Enqueued Time	Delivery Count	State	Body ...	Label/Subject	Message Text
Peek or receive messages to view them here.							

7. Type a message and click on Send. To see your messages, you need to click on Peek next message and you will see your message respectively.

Send messages

Content type

text/plain

Upload .txt file ⓘ

Select file

Browse

Message Body

1 Testing messages

Peek Mode

Send messages

Refresh

Export messages

Show message body

Settings

Learn more

Give feedback

Queue (1)

Dead-letter (0)

Peek from start

Peek next messages

Peek with options

Re-send selected messages

Download selected message body

Showing 1 of 1 messages

☒	Sequence Number	Message ID	Enqueued Time	Delivery Count	State	Body ...	Label/Subject	Message Text
☒	1	c07dd6fed2f64e78acd955e138c3...	Wed, Mar 12, 25, 03:56:45 PM G...	0	Active	16 B		Testing messages

Message Body

Message Properties

Fit message body

Off

Testing messages

8. Once you are done just delete this queue in your Service bus and move to the next lab.